

XIAOSHENG ZHAO

xzhao113@jh.edu

+1 443-925-0076 ◇ Baltimore, USA

RESEARCH INTERESTS

My research focuses on using machine learning (ML) to advance astronomy—using ML to uncover the Milky Way’s composition and evolutionary history, to probe the universe’s origin and content, and to bridge these scales. As a member of the Subaru Prime Focus Spectrograph (PFS) JHU Galactic Archaeology group, I develop and apply ML methods, including foundation models, to stellar spectra for chemical abundance and radial velocity estimation, and to constrain the inner dark matter profiles of dwarf spheroidal galaxies, maximizing the scientific yield of PFS.

EDUCATION

Tsinghua University <i>PhD in Astronomy</i> <i>Advisor: Prof. Yi Mao</i> <i>Thesis title: Explore the epoch of reionization with machine learning</i>	Sep 2018 - Jun 2024
Wuhan University <i>BS in Physics</i>	Sep 2014 - Jun 2018

PROFESSIONAL POSITION

Johns Hopkins University <i>Assistant Research Scientist</i> <i>Working with: Prof. Alex S. Szalay & Prof. Rosemary F.G. Wyse</i>	Nov 2025 - Now
Johns Hopkins University <i>Postdoctoral Fellow, focus—maximizing science return from PFS</i> <i>Mentor: Prof. Alex S. Szalay & Prof. Rosemary F.G. Wyse</i>	Aug 2024 - Nov 2025

RESEARCH EXPERIENCE

University of Chinese Academy of Sciences/NAOC <i>Short-term visiting, focus—stellar spectra foundation models.</i> <i>Host: Prof. Yang Huang</i>	Jul 2024 - Aug 2024
Institut d’Astrophysique de Paris <i>Long-term visiting, focus—(explainable) ML for astrophysics and cosmology</i> <i>Mentor: Prof. Benjamin D. Wandelt</i>	Nov 2022 - May 2024

PUBLICATION

Refereed Journals

[Generalization from Low- to Moderate-Resolution Spectra with Neural Networks for Stellar Parameter Estimation: A Case Study with DESI](#)

Xiaosheng Zhao, Yuan-Sen Ting, Rosemary F.G. Wyse, Alexander S. Szalay, Yang Huang, László Dobos, Tamás Budavári, Viska Wei
2026, ApJ, 1006, 2

[SpecCLIP: Aligning and Translating Spectroscopic Measurements for Stars.](#)

Xiaosheng Zhao; Yang Huang; Guirong Xue; Xiao Kong; Jifeng Liu; Xiaoyu Tang; Timothy C. Beers; Yuan-Sen Ting; A-Li Luo
2026, ApJ, 998, 189

[Ordered embeddings and intrinsic dimensionalities with information-ordered bottlenecks.](#)

Matthew Ho; **Xiaosheng Zhao**; Benjamin D. Wandelt
2025, Machine Learning: Science and Technology (MLST)

[Simulation-based Inference of Reionization Parameters from 3D Tomographic 21 cm Light-cone Images. II. Application of Solid Harmonic Wavelet Scattering Transform.](#)

Xiaosheng Zhao; Yi Mao; Shifan Zuo; Benjamin D. Wandelt
2024, ApJ, 973, 41

[Can Diffusion Model Conditionally Generate Astrophysical Images?](#)

Xiaosheng Zhao; Yuan-Sen Ting; Kangning Diao; Yi Mao
2023, MNRAS, 526, 1699

[Implicit Likelihood Inference of Reionization Parameters from the 21 cm Power Spectrum](#)

Xiaosheng Zhao; Yi Mao; Benjamin D. Wandelt
2022, ApJ, 933, 236

[Simulation-Based Inference of Reionization Parameters From 3D Tomographic 21 cm Lightcone Images.](#)

Xiaosheng Zhao; Yi Mao; Cheng Cheng ; Benjamin D. Wandelt
2022, ApJ, 926, 151

[How to evaluate the sufficiency and complementarity of summary statistics for cosmic fields: an information-theoretic perspective](#)

Ce Sui; Yi Mao; **Xiaosheng Zhao**; Tao Jing; Benjamin D. Wandelt
2026, ApJ, 998, 2

[Square Kilometre Array Science Data Challenge 3a: foreground removal for an EoR experiment.](#)

A. Bonaldi; ...; **Xiaosheng Zhao**; ...
2025, MNRAS, 543, 2

Refereed Conference proceedings

[Finetuning Stellar Spectra Foundation Models with LoRA](#)

Xiaosheng Zhao; Yuan-Sen Ting; Alexander S. Szalay; Yang Huang
2025, ICML 2025 Workshop on Machine Learning for Astrophysics

[3D ScatterNet: Inference from 21 cm Light-cones](#)

Xiaosheng Zhao; Shifan Zuo; Yi Mao
2023, ICML 2023 Workshop on Machine Learning for Astrophysics

[Evaluating Summary Statistics with Mutual Information for Cosmological Inference.](#)

Ce Sui; **Xiaosheng Zhao**; Tao Jing; Yi Mao
2023, ICML 2023 Workshop on Machine Learning for Astrophysics

Under review

Foundation Models for Astrophysics

Xiaosheng Zhao; Yuan-Sen Ting

Invited chapter for the edited book “Machine Learning Techniques for Astrophysics and Cosmology”

Machine Learning and the SKA for Cosmic Dawn and the Epoch of Reionization

Anshuman Acharya; ...; **Xiaosheng Zhao**

Published in Advancing Astrophysics with the SKA II (AASKAII)

Galactic Archaeology with the Subaru ‘Ōnohi’ula Prime Focus Spectrograph Strategic Program

Masashi Chiba; ...; **Xiaosheng Zhao**

Likelihood-free Model Selection in Cosmic Reionization with Three-dimensional Tomographic 21 cm Lightcone Images.

T Binnie; **Xiaosheng Zhao**; JR Pritchard; Yi Mao

2025, Submitted to AAS Journals

TALKS & PRESENTATIONS (SELECTED)

2nd Annual Cosmic Horizons Conference

Jul 2026

Contributed talk: *Decoding the Chemical Fossil Record: Machine Learning and Foundation Models for Near-Field Cosmology*
UVA, USA

LSST Discovery Alliance Regional Meeting

May 2026

Contributed talk: *Cross-Modal Learning for Galactic Archaeology: Prospects in the Rubin Era* JHU, USA

BCCP/Cosmology seminar

Apr 2026

Seminar: *Decoding the Chemical Fossil Record: Machine Learning and Foundation Models for Near-Field Cosmology*
UC Berkeley, USA

Center for Decoding the Universe Journal Club

Apr 2026

Journal Club: *Decoding the Chemical Fossil Record: Machine Learning and Foundation Models for Near-Field Cosmology*
Stanford, USA

247th AAS Meeting

Jan 2026

Contributed talk: *SpecCLIP: Aligning and Translating Spectroscopic Measurements for Stars* Phoenix, USA

ICML2025 ML4Astro

Jul 2025

Spotlight talk: *Finetuning Stellar Spectra Foundation Models with LoRA* Vancouver, Canada

JHU/STScI CAS wine & cheese seminar

Apr 2025

Seminars: *From 21 cm Astrophysics to Galactic Archaeology: Enriching Physics-Driven Analysis with Machine Learning*
JHU, USA

Astro Coffee

Sept 2023

Informal talk: *Can Diffusion Model Conditionally Generate Astrophysical Images?* IAS, USA

Understanding the epoch of reionization	Mar 2023
Contributed talk: <i>Implicit Likelihood Inference of Reionization Parameters from 21 cm Power Spectrum and solid harmonic wavelet scattering coefficients</i>	<i>Sexten, Italy</i>
SAZERAC 21cm 2022	Mar 2022
Contributed talk: <i>Implicit Likelihood Inference of Reionization Parameters from the 21 cm Power Spectrum</i>	<i>Virtual</i>
SAZERAC SIP, learning the high-redshift universe	Feb 2022
Contributed talk: <i>Simulation Based Inference of Reionization Parameters From 3D Tomographic 21 cm Lightcone Images</i>	<i>Virtual</i>
SKA CD/EoR Science Telecon	Jul 2021
Contributed talk: <i>Simulation Based Inference of Reionization Parameters From 3D Tomographic 21 cm Images</i>	<i>Virtual</i>
HERA telecon	Jun 2021
Invited talk: <i>Simulation Based Inference of Reionization Parameters From 3D Tomographic 21 cm Images</i>	<i>UC, Berkeley, USA</i>

OUTREACH & SERVICE

Reviewer for ApJ, MNRAS, A&A, JCAP, RAA, and JATIS.

MENTORING & TEACHING EXPERIENCE

Teaching Assistant in undergraduate *Physics* course. Feb - Jun 2019

SKILLS

Coding languages: {Python, Jax} (Fluent), {C, Shell, html&CSS}(Basic)

General: Astrophysical data science and machine learning application with Pytorch, Pandas, Scikit-learn, Tensorflow, particularly for astronomy.

AWARDS

Comprehensive Scholarship (First-class) (<i>University-level scholarship</i>)	2021 - 2022
AMD Scholarship (<i>Top scholarship awarded to two graduate students in the department this year</i>)	2020 - 2021
Future Scholar Scholarship of Tsinghua University (<i>University-level scholarship</i>)	2018
National Scholarship	2015 - 2016